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RESEARCH ARTICLE

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An automatic monitoring method for disturbance caused by deforestation and forest fires

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ABSTRACT

Timely and accurate detection and assessment of forest disturbances are crucial for early warning and risk prevention of forest disasters. This study presents an automated forest disturbance monitoring algorithm with three main steps: (1) extracting features that capture the remote sensing response of forest fires and deforestation and creating threshold maps of features; (2) using the threshold maps to extract potential forest disturbance pixels as training samples to train a one-class support vector machine (OC-SVM) and detect forest disturbance regions; and (3) applying diagnostic rules to diagnose the disturbance type as either burned or deforested areas. We validated the method using 10 global cases (five forest fires and five deforestation events). The results show that Sentinel-2 bands 6, 7, 8 and 8A, the mean textures of bands 6, 7, 8 and 8A and the difference vegetation index (DVI) are effective for disturbance detection. The overall detection accuracy of the 10 cases is 98.46% on average, with an average F1 score of 0.923 and a type diagnosis accuracy of 100%. This approach enables high-precision, automated detection and classification of forest disturbances and can be extended to monitor other disturbance types, offering a valuable tool for forest disaster response and management.

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KEYWORDS
Forest disturbances; automatic detection; comprehensive disturbance features

1. Introduction

Forests, as integral components of terrestrial ecosystems, play crucial roles in regulating climate, pollution control, and flood regulation (Cao et al. 2022; Aziz et al. 2024). Disturbances are inherent components of forest ecosystems; they are characterised by randomness and destructiveness and typically occur as discrete events (Dale et al. 2000; Seidl et al. 2011). Forest disturbances can be divided into two categories according to their cause: natural disturbances and human disturbances caused by biological or abiotic factors (Adil et al. 2022). The increased frequency of forest disturbance events, driven by human activities and extreme climate conditions, has significantly affected the dynamic growth of forest vegetation. Forest disturbances lead to a reduction in forest biomass, function, and species diversity. Concurrently, the oxidation and decomposition of wood release carbon into the atmosphere, further impacting the provision of ecosystem services by forests (McDowell et al. 2020; Ye et al. 2021; Zhu et al. 2023). Over the past decade, the global net loss of forests has occurred at a rate of 4.7 million hectares annually (FAO 2020). If current trends continue, the global potential canopy cover could decrease by approximately 223 million hectares by 2050 (Bastin et al. 2019). The main factors affecting forest disturbances vary from region to region. Deforestation is prominent in tropical regions, especially in Southeast Asia and Latin America, whereas in temperate regions, other human disturbances, such as forest fires, play a major role(Curtis et al. 2018; Altmann et al. 2020). Therefore, accurately and promptly collecting data on the extent of forest disturbances, particularly concerning forest fires and deforestation, is crucial for implementing early warning systems and effective management to prevent further loss of forested areas.

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Remote sensing serves as a crucial approach for forest disturbance monitoring, involving two key processes: extracting disturbance features and developing effective monitoring methods (Rodríguez Paulino et al. 2024; Fassnacht et al. 2024). When forests experience disturbances, features such as spectral, scattering, radiometric, and textural features in the affected regions undergo significant changes (Xiao, Wang, and Zhang 2024). Disturbance feature extraction, which focuses on extracting specific remote sensing bands or combining them to create indices that effectively highlight forest disturbances, is pivotal in forest disturbance monitoring (Li et al. 2022; Du et al. 2023; Coops et al. n.d.). In forest fire monitoring, indices such as the normalised burn ratio (NBR) are widely used. A study by Hislop demonstrated that both the normalised difference vegetation index (NDVI) and NBR effectively distinguish burned areas from unburned areas (Hislop et al. 2019). For deforestation monitoring, the NDVI was employed as a metric to assess the temporal dynamics of vegetation (Hamunyela, Verbesselt, and Herold 2016). Zhang et al. enhanced small-scale tropical disturbance monitoring by fusing Landsat and Sentinel-2 self-referenced normalised burn ratio (rNBR) images using a deep learning-based downscaling method to generate high-resolution (10 m) annual rNBR images (Zhang et al. 2021). Schroeder's study highlights the effectiveness of shortwave infra-red (SWIR) bands in detecting forest fires and deforestation (Schroeder et al. 2011). Additionally, Bonannella et al. analysed Landsat-8 bands and spectral indices for monitoring forest disturbances and reported that Landsat-8 band 6 (SWIR1) was most effective for detecting deforestation, Landsat-8 band 7 (SWIR2) was optimal for monitoring forest fire disturbances, and NBR led to consistent performance across both scenarios (Bonannella et al. 2022).

Disturbance monitoring methods include threshold, traditional machine learning, and deep learning approaches. Hu et al. used Sentinel-2 bands 4, 11, and 12 with a multithreshold method for fire detection (Hu, Ban, and Nascetti 2021). However, owing to the differences in vegetation types and environments in different regions, the threshold determined in a specific area cannot be directly transferred to other areas. Moreover, the threshold designed for a specific sensor may not be applicable to other sensors because of the difference in different sensor band settings (Hirschmugl et al. 2017). The machine learning models commonly used for forest disturbance detection include support vector machines (SVM) (Li and xu 2010; Cao et al. 2014), decision trees (DT) (Im and Jensen 2005), random forests (RF) (Seo et al. 2017), etc. Mohammadpour et al. used Sentinel-2 data to calculate the principal components of 10 spectral bands, along with corresponding texture features and vegetation indices such as the NDVI, green normalised difference vegetation index (GNDVI), enhanced vegetation index (EVI), and soil-adjusted vegetation index (SAVI), and demonstrated that incorporating at least one texture feature and one vegetation index with Sentinel-2 spectral bands significantly improved forest monitoring accuracy when the RF algorithm was applied (Mohammadpour, Viegas, and Viegas 2022). Huang et al. analysed changes in forest cover caused by clearcutting, wildfires, and other factors using the NDVI and the SVM method (Huang et al. 2008). Commonly used deep learning models for forest disturbance detection include convolutional neural networks (CNN) and U-Net models. Bragagnolo et al. used Sentinel-2 spectral bands 4, 3, 2, and 8 to generate a deforestation map of the Brazilian tropical rainforest by integrating the U-Net fully convolutional neural network architecture with a logistic growth model (Bragagnolo, da Silva, and Grzybowski 2021). Solórzano et al. employed the U-Net 3D and SVM algorithms, which used Sentinel-1 VV and VH polarisations along with Sentinel-2 blue (B), green (G), red (R), and near-infra-red (NIR) spectral bands, to detect deforestation (Solorzano et al. 2023). Almeida et al. monitored wildfires in the Amazon rainforest and issued wildfire alerts using RGB images and a CNN model (Almeida et al. 2022). Overall, existing features have typically been developed to address specific types of forest disturbances, with a focus on detecting the extent and intensity of known disturbances. However, these methods are insufficient for real-time monitoring of unknown forest disturbances. Real-time monitoring of unknown forest disturbances involves the rapid detection of disturbance events as they occur, along with the determination of their types. This capability allows for immediate monitoring of forest conditions in targeted areas, providing crucial, timely information for effective forest protection and management.

Compared with traditional forest disturbance monitoring, real-time detection and diagnosis of unknown disturbances face greater challenges. It requires remote sensing features that can capture multiple disturbance types, lightweight algorithms for timely processing, and accurate identification of specific disturbance types such as fires and deforestation. To address these challenges, this study focuses on the two most common types of forest disturbance—fire and deforestation. Sentinel-2 imagery was selected

as the primary data source. Although sensors such as MODIS offer higher temporal resolution, their coarse spatial resolution limits the detection of small or fragmented disturbances. Sentinel-2, with its 10 m resolution and red-edge and shortwave infra-red bands, provides superior capability for precisely delineating disturbance areas and differentiating between burned and deforested regions, while still supporting near-real-time monitoring through automated processing.

Accordingly, this study selects remote sensing features suitable for both fire and deforestation detection and develops automated methods to detect the spatial extent of forest disturbances and diagnose disturbance types (burned or deforested areas).

2. Forest disturbance areas and datasets

2.1. Study cases

To validate our designed forest disturbance detection and diagnostic method, we selected 10 research cases from different regions around the world. The distribution of 10 cases is shown in Figure 1, and the detailed information of each case is shown in Table 1. These selected areas span a wide range of climate zones and forest ecosystems, reflecting the global heterogeneity of forest environments. The fire-affected regions include Potenza, Italy, which experiences an oceanic climate and is dominated by deciduous broad-leaved forests; Xichang, China, with a subtropical monsoon climate and evergreen coniferous–broadleaved mixed forests; British Columbia, Canada, characterised by a temperate oceanic climate and temperate evergreen coniferous forests; President Hayes, Paraguay, with a dry tropical forest climate and deciduous broad-

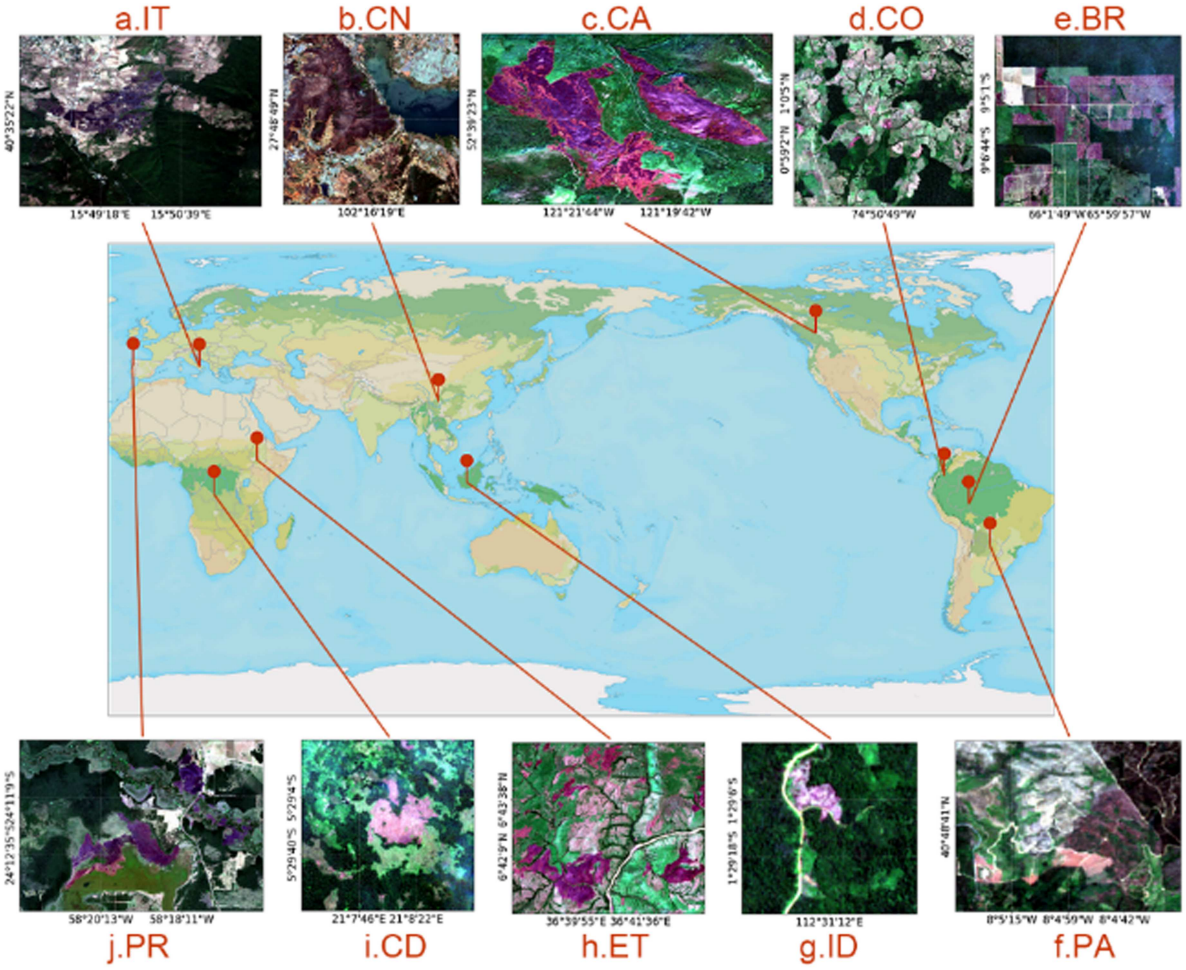


Figure 1. Distribution of 10 cases of forest disturbance.

Table 1. Detailed information on 10 disturbance cases.

Cases	Location	Abbreviation	Occurrence date	Disturbance types
(a)	Potenza, Italy	IT	August 2017	Burned area
(b)	Xichang City, China	CN	April 2020	Burned area
(c)	British Columbia, Canada	CA	August 2024	Burned area
(d)	Caquetá, Colombia	CO	August 2023	Deforestation
(e)	Amazonas, Brazil	BR	September 2022	Deforestation
(f)	Presidente Hayes, Paraguay	PA	July 2019	Deforestation
(g)	central Kalimantan, Indonesia	ID	April 2023	Deforestation
(h)	the border between South West Ethiopia and South Ethiopia.	ET	April 2020	Burned area
(i)	Kasaï Province, Democratic Republic of the Congo	CD	April 2020	Deforestation
(j)	Viseu, Portugal	PR	March 2020	Burned area

leaved vegetation; and the border between southwestern and southern Ethiopia, featuring a tropical monsoon climate with dry evergreen montane forests and grassland mosaics. The deforestation cases also span a variety of tropical forest environments. Caquetá, Colombia, Amazonas, Brazil, Central Kalimantan, Indonesia, and Kasaï, Democratic Republic of the Congo all lie within humid equatorial zones dominated by tropical evergreen broadleaf forests, under tropical rainforest climates. In contrast, Viseu, Portugal represents a temperate forest environment with a Mediterranean climate and broad-leaved forest vegetation. This geographically and ecologically diverse case selection captures a wide spectrum of disturbance scenarios, climatic gradients, and forest structures, allowing for a comprehensive evaluation of the detection algorithm under various real-world conditions.

2.2. Datasets

The Sentinel-2 satellite launched by the European Space Agency (ESA) provides multispectral high-resolution optical images of the global land surface with a resolution of 10 m at a revisit time of 5 days, covering 13 bands from VIS and NIR to SWIR. Notably, the unique red-edge band is very sensitive to the chlorophyll content of vegetation, providing rich remote sensing information for monitoring forest disturbances. In this study, we obtained time series of Sentinel-2 MSI Level-2A images of the study areas for a period of one year from the Google Earth Engine (GEE) platform. To ensure data quality, we limited the degree of cloud coverage to under 20%, used quality assessment band QA60 to remove clouds from each image and resampled all the bands to a 10 m resolution. The ee.ImageCollection.median() function was subsequently used to calculate the median image for each band of images to obtain a monthly composites of Sentinel-2 images.

3. Automated forest disturbance monitoring method

This study introduces a pixel-scale automated detection and diagnostic algorithm for monitoring forest fire and deforestation scenarios (Figure 2). The methodology involves the following key steps: (1) extracting comprehensive remote sensing response features that effectively capture forest fires and deforestation, (2) calculating pixel-scale change rates on the basis of the selected features and creating a threshold map for each selected feature, (3) using the threshold maps to extract potential forest disturbance pixels as training samples for further classification, (4) employing OC-SVM to detect forest disturbance regions, and (5) applying diagnostic rules to diagnose the detected disturbance regions as either burned areas or deforested areas.

3.1. Comprehensive feature extraction for multitype forest disturbance monitoring

For remote sensing feature selection, Italian fires and Brazil deforestation were selected as research cases, and Sentinel-2 remote sensing images before and after forest disturbance were obtained from the GEE.

In each case, 200 disturbance samples and 200 nondisturbance samples were manually selected through visual interpretation. The separability of different remote sensing features between disturbance and nondisturbance samples was calculated and compared to screen remote sensing response features that are suitable for detecting both forest fires and deforestation.

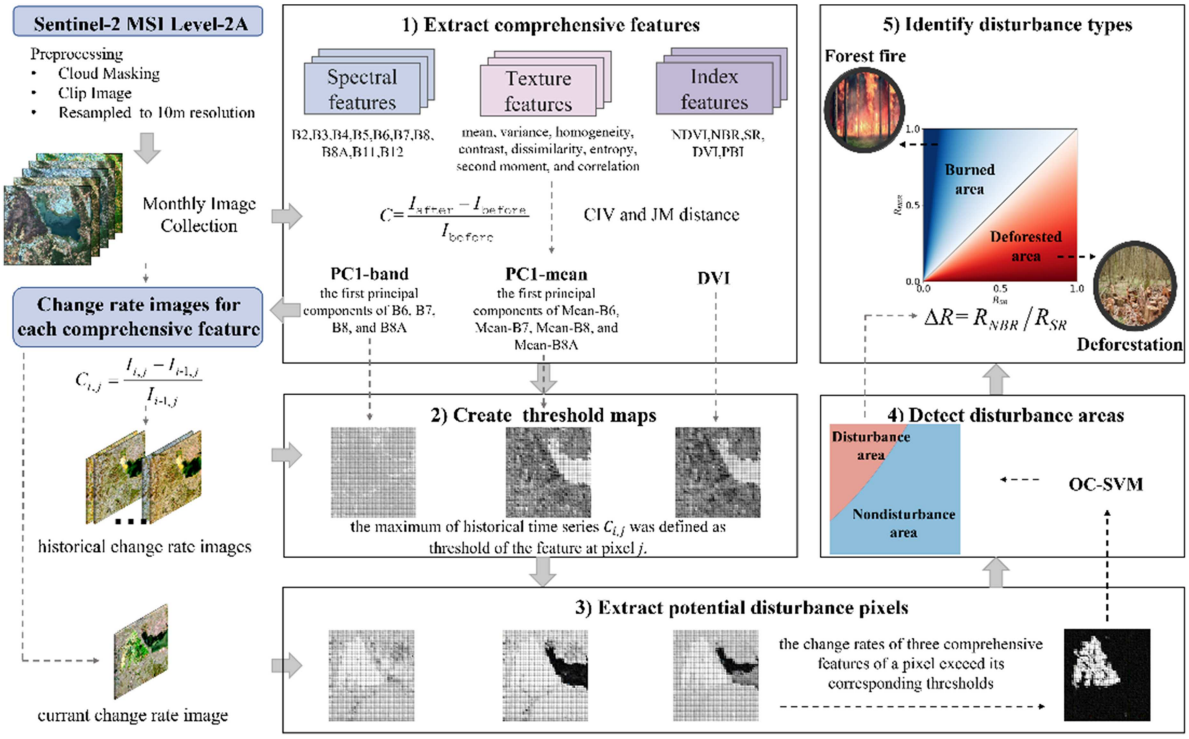


Figure 2. Technical flowchart.

Table 2. Specific information about the selected features.

Feature type	Name	Description	References
Spectral features	B2	Blue	
	B3	Green	
	B4	Red	
	B5	Red Edge 1	
	B6	Red Edge 2	
	B7	Red Edge 3	
	B8	NIR	
	B8A	Red Edge 4	
	B11	SWIR 1	
	B12	SWIR 2	
Texture features	Mean	$Mean = \sum_i \sum_j P(i,j) \cdot i$	(Haralick, Shanmugam, and Dinstein 1973)
	Variance	$Variance = \sum_i \sum_j P(i,j) \cdot (i - Mean)^2$	(Haralick, Shanmugam, and Dinstein 1973)
	Homogeneity	$Homogeneity = \sum_i \sum_j P(i,j) \cdot \frac{1}{1+(i-j)^2}$	(Haralick, Shanmugam, and Dinstein 1973)
	Contrast	$Contrast = \sum_i \sum_j P(i,j) \cdot (i-j)^2$	(Haralick, Shanmugam, and Dinstein 1973)
	Dissimilarity	$Dissmilarity = \sum_i \sum_j P(i,j) \cdot \lvert i-j \rvert$	(Haralick, Shanmugam, and Dinstein 1973)
	Entropy	$Entropy = \sum_i \sum_j P(i,j) \cdot lnP(i,j)$	(Haralick, Shanmugam, and Dinstein 1973)
	Second Moment	$Second\ Moment = \sum_i \sum_j P(i,j)^2$	(Haralick, Shanmugam, and Dinstein 1973)
	Correlation	$Correlation = \sum_i \sum_j \frac{(i-Mean)\cdot(j-Mean)\cdot P(i,j)^2}{Variance}$	(Haralick, Shanmugam, and Dinstein 1973)
Index features	NDVI	$NDVI = (NIR - Red)/(NIR + Red)$	(Navarro et al. 2017)
	NBR	$NBR = (NIR - SWIR1)/(NIR + SWIR1)$	(Veraverbeke et al. 2012)
	Simple ratio index (SR)	$SR = NIR/Red$	(Birth and McVey 1968)
	Difference vegetation index (DVI)	$DVI = NIR - Red$	(Richardson 1977)
	Plant biochemical index (PBI)	$PBI = NIR/Green$	(Nidamanuri et al. 2008)

The remote sensing features involved in the analysis are the change rates (C) before and after forest disturbance of a total of 95 remote sensing features, including reflectance bands, textures, and indices listed in Table 2. C is calculated as follows:

$$C = \frac{I_{\text{after}} - I_{\text{before}}}{I_{\text{before}}} \tag{1}$$

where I_{after} represents the feature value of a certain pixel after forest disturbance. I_{before} represents the median feature value of the pixel in the most recent month before forest disturbance occurred. C highlights the change information of the disturbed part and suppresses the change information of the undisturbed part.

Spectral features are the direct basis for judging and classifying the types of objects in remote sensing images. The B2-B8A, B11, and B12 bands of the Sentinel-2 image were selected as the spectral features. Among them, the B5-B7 bands are the red edge bands unique to Sentinel-2.

Texture features provide information about the structural arrangement of an object and its relationship to its local neighbourhood. The grey-level cooccurrence matrix (GLCM) is a second-order statistic that describes the spatial structural features and correlations between pixel pairs through the spatial relationships of image grey values. In this study, 8 texture features are calculated under a window size of 3×3. These features include the mean, variance, homogeneity, contrast, dissimilarity, entropy, second moment, and correlation.

The NDVI is the most commonly used vegetation index in remote sensing images and is widely regarded as the best indicator for assessing vegetation growth status and distribution density. The NBR is mainly used to extract information from burned areas and monitor the recovery of vegetation in burned areas. After vegetation is affected by fire, the reflectivity of the near-infra-red band decreases, whereas the reflectivity of the shortwave infra-red band remains relatively high. Areas with lower NBR values usually indicate areas with more severe fire damage. The SR is mainly used to assess the status and health of vegetation. A higher SR usually indicates lush and healthy vegetation, whereas a lower SR indicates that the vegetation may be under pressure or in poor condition. The DVI is highly sensitive to soil background changes and has a good monitoring effect for areas with low vegetation coverage or early and middle stages of vegetation development. The PBI is often used to quantify the growth status and biomass accumulation of vegetation, which helps monitor vegetation growth, drought, land degradation, etc.

The indicators used for the separability of different remote sensing features include the Jeffries–Matusita (JM) distance and common isolation value (CIV). The JM distance assumes that the data follow a multivariate normal distribution, which is expressed as follows:

$$JM = 2(1 - e^{-B}) \tag{2}$$

$$B = \frac{1}{8}(\mu_1 - \mu_2)^2 \frac{2}{\sigma_1^2 + \sigma_2^2} + \frac{1}{2}\ln\left(\frac{\sigma_1^2 + \sigma_1^2}{2\sigma_1\sigma_2}\right) \tag{3}$$

where μ_1 and μ_2 represent the means of disturbance and nondisturbance samples, and σ_1^2 and σ_2^2 represent the variances of disturbance and nondisturbance samples, respectively. The JM values range between 0 and 2, with values exceeding 1 indicating good separability between the two classes. The higher the JM value is, the better the separability.

We also used the isolation forest (IF) to evaluate the separability between disturbance and nondisturbance samples (Valdivieso-Ros, Alonso-Sarria, and Gomariz-Castillo 2021). This method does not require the data to follow a normal distribution. The core idea is to train an IF model on one class of data (class A, such as disturbance samples) and then measure how similar other data (class B, such as nondisturbance samples) are to that class (class A). The CIV is then defined as the ratio of the intersection to the union of the two classes (Figure 3). The higher the CIV value is, the lower the separability between classes.

An important characteristic of the CIV measure is its asymmetry; that is, *CIVA, B* is not equal to *CIVB, A*. When the isolation of classes A and B is computed using an IF trained using class B, the CIV reflects the omission error risk for class A and the commission error risk for class B. Conversely, when the isolation is computed using an IF trained using class A, the CIV relates to the omission error risk for class B and the commission error risk for class A. To accurately detect the forest disturbance class, we should train the IF using the forest disturbance class. This approach allows the model to maximise its ability to detect forest disturbances, thereby reducing the risk of mistakenly detecting them as normal categories and improving detection accuracy.

The CIV and JM distance results are not completely consistent. Previous studies have demonstrated that the CIV reflects more information than the JM distance does (Alonso-Sarria, Valdivieso-Ros, and

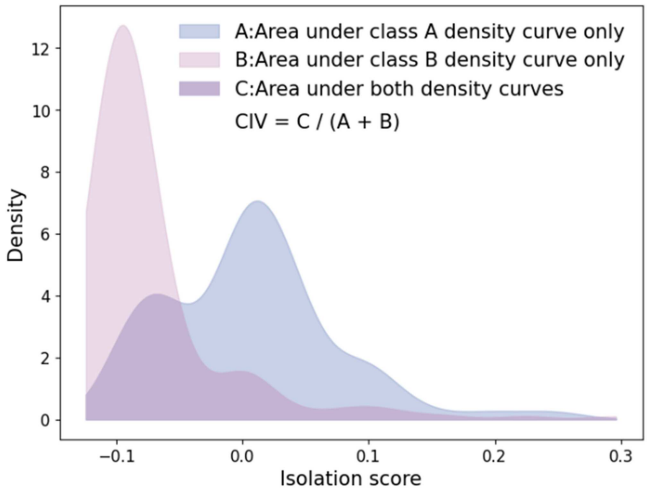


Figure 3. Calculation of the CIV.

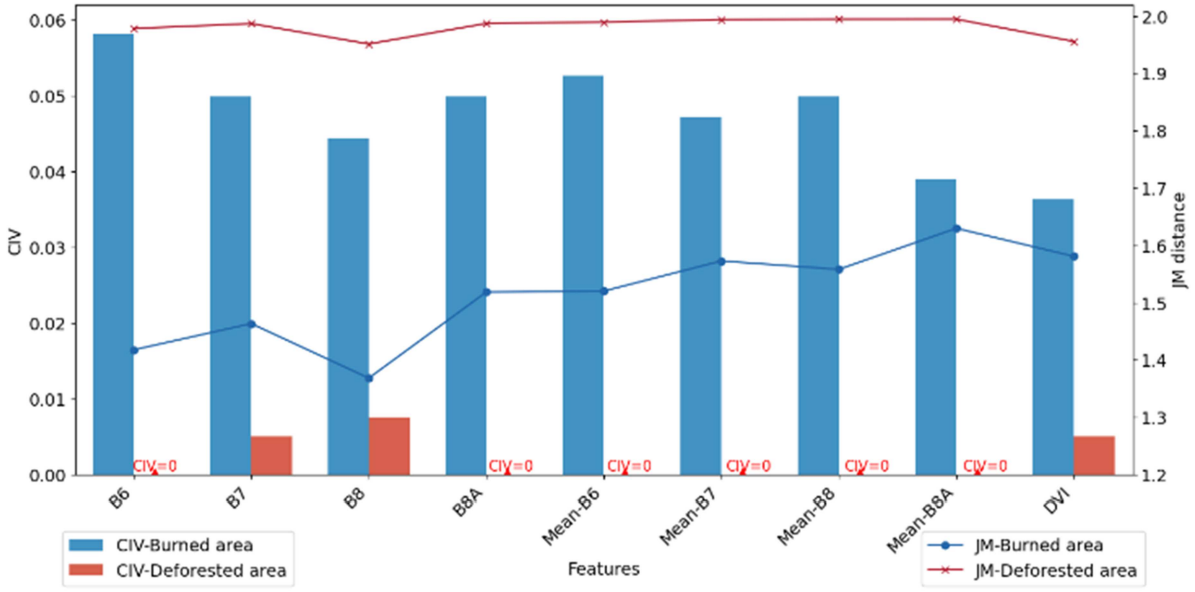


Figure 4. Feature separability analysis of Italy forest fire and Brazil deforestation based on CIV and JM distance.

Gomariz-Castillo 2019). In addition, the JM relies on the assumption of a normal distribution, which may not be applicable to these feature datasets. Therefore, the CIV method is prioritised during feature selection, with the JM distance used as a reference. Specifically, we first selected features with a CIV index less than 0.1 in both cases of fire and deforestation as candidate features. The JM distance of candidate features in Italian fires is generally greater than that in Brazilian deforestation. The candidate features with JM distances greater than 1.0 for Italian fires and greater than 1.5 for Brazilian deforestation were further selected as the final features. The selected features are shown in Figure 4. These features include B6, B7, B8, B8A, Mean-B6, Mean-B7, Mean-B8, Mean-B8A, and DVI. To reduce feature dimensionality, principal component analysis (PCA) was applied to compress features of the same type. Specifically, the first principal components of B6, B7, B8, B8A (PC1-band), Mean-B6, Mean-B7, Mean-B8, and Mean-B8A (PC1-mean), in addition to the DVI, were ultimately used as comprehensive features for the detection of forest disturbances.

The selection of these spectral and spatial features is grounded in their well-established physical significance for forest disturbance monitoring. Specifically, red-edge and near-infra-red bands (B6–B8A)

exhibit high sensitivity to vegetation chlorophyll content and canopy structure, key indicators of vegetation health and stress. Texture features capture spatial heterogeneity within the imagery, directly reflecting patterns of canopy disruption and fragmentation associated with disturbances. Furthermore, the Difference Vegetation Index (DVI) enhances spectral contrast between disturbed areas and undisturbed, healthy vegetation. Collectively, these features provide complementary and robust indicators, rendering them highly feasible and rational for effectively detecting and characterising forest disturbance dynamics.

3.2. Forest disturbance detection and diagnosis technology process based on comprehensive features

3.2.1. Creation of feature change rate threshold maps

On the basis of the preprocessed monthly composites of Sentinel-2 images, for a given pixel j, we used formula 4 to calculate the change rate (C) of features (I) in adjacent months (i, $i \leq 12$) and obtain the time series $C_{i,j}$. Then, for each comprehensive feature selected, the maximum of $C_{i,j}$ was calculated pixel-by-pixel to represent the maximum historical change (i.e. threshold) of the feature at pixel j. As a result, three threshold maps

$$C_{i,j} = \frac{I_{i,j} - I_{i-1,j}}{I_{i-1,j}} \tag{4}$$

corresponding to the three comprehensive features were obtained.

3.2.2. Extraction of potential disturbance pixels on the basis of change rate threshold maps

For each pixel, the current change rate image (C) is compared against the threshold maps for each comprehensive feature (Figure 5). If the change rates of three comprehensive features of a pixel exceed its corresponding thresholds, the pixel will be marked as a potential disturbance pixel. If the potential disturbance pixel is an isolated pixel, we will relabel it as 'other' pixel.

3.2.3. Potentially disturbance pixels drive OC-SVM classification to detect disturbance areas

The one-class support vector machine (OC-SVM) is a specialised form of support vector machine that requires only samples from a single target class to separate it from a complex background and can be considered as a conventional binary support vector machine, where all training data belong to the first class, and the origin serves as the sole member of the second class (Li and xu 2010). The OC-SVM maps the input data of the single-class samples into a high-dimensional feature space using a kernel function. Through iterative optimisation, the maximum-margin hyperplane that separates the training data from the origin is identified. The equation for the separating hyperplane is given by:

$$f(x) = <x, w> + b \tag{5}$$

where w is the normal vector and b is the bias term. If $f(x) > 0$, then x belongs to the target class; otherwise, it is classified as an outlier. This approach enables the detection of the target class. This method requires only a single class of input samples for classification, reducing the dependence on samples from different target classes.

By using kernel functions, the optimisation problem of the OC-SVM is transformed into an equivalent dual quadratic programming problem.

$$\min_{\alpha} \frac{1}{2} \sum_{i,j} \alpha_i \alpha_j K(x_i, x_j) \tag{6}$$

subject to $0 \leq \alpha_i \leq \frac{1}{vl}$, and $\sum_i \alpha_i = 1$, where α_i is a Lagrange multiplier, representing the weight of the example x_i. Vectors with nonzero weights are referred to as 'support vectors' and are the only vectors that determine the optimal hyperplane. The parameter v controls the trade-off between maximising the number

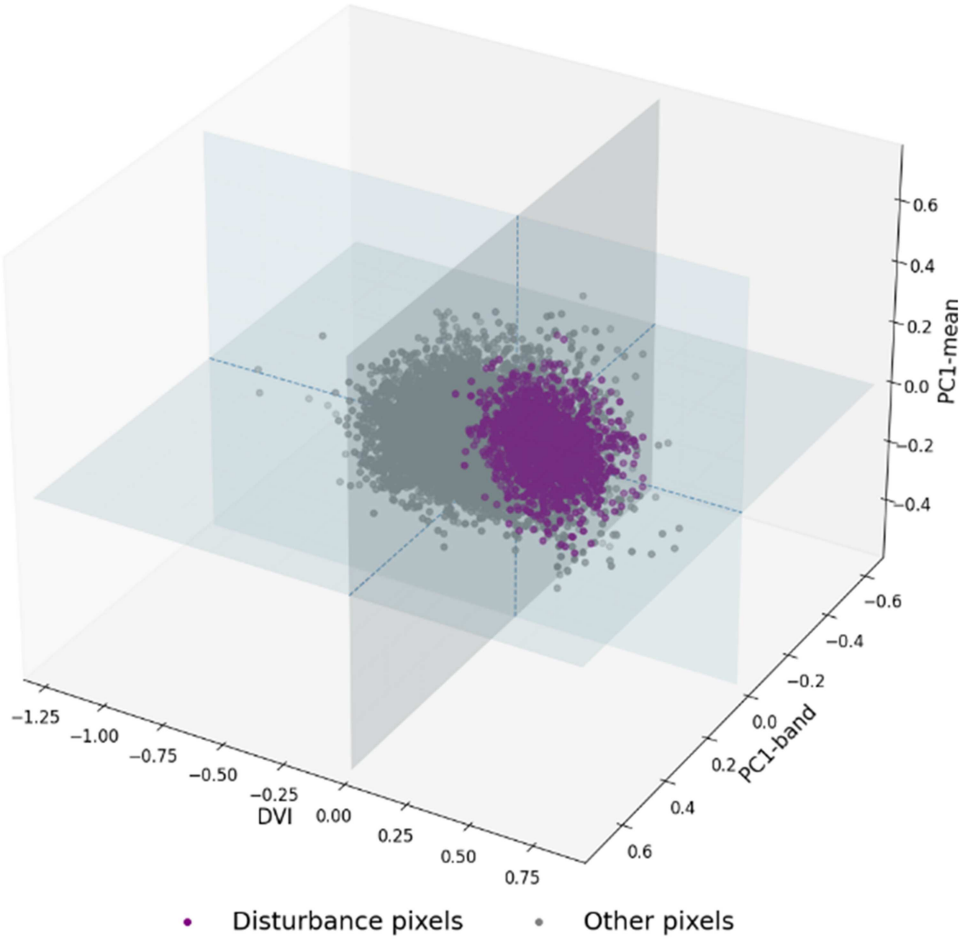


Figure 5. The difference values between the current change rate image and the historical threshold are calculated by subtracting the historical threshold from the current change rate. The x-axes, y-axes, and z-represent three comprehensive feature bands. In the purple region, all three types of feature vectors are greater than zero, indicating that the current change rates exceed the historical thresholds and are thus marked as disturbance samples.

of data points contained by the hyperplane and the distance of the hyperplane from the origin. The parameter l represents the number of points in the training dataset, and $K(x_i, x_j)$ is the kernel function.

By using a kernel function to project input vectors into a feature space, OC-SVM can accommodate nonlinear decision boundaries. In this study, we use the radial basis function (RBF) kernel, which is a common choice for OC-SVM because of its flexibility in handling various data distributions.

$$K(x_i, x_j) = \exp(-\gamma \|x_i - x_j\|^2) \tag{7}$$

where γ is a kernel parameter to be set for a specific problem.

Notably, OC-SVM is parameterised by $\nu \in (0, 1]$ and the parameter $\gamma \in (0, \infty)$ within the selected RBF kernel, both of which directly influence classification accuracy. To determine the optimal parameters, a randomised grid search is used to test a finite set of parameter combinations and select those that yield the highest accuracy. The sklearn.svm.OneClassSVM() function is used for classification. ν and γ correspond to the parameters nu and gamma, respectively. The search spaces for these parameters are defined as 'nu': [0.01,0.1, 0.3, 0.5, 0.7, 0.9] and 'gamma': [0.01, 0.1, 1, 5, 10, 100], respectively. The parameter selection results, obtained through a 7-fold cross-validation process, enable efficient parameter tuning designed for the data. The input for the OC-SVM model consists of the current change rate image (C) and the disturbance pixels extracted to detect disturbance areas.

3.2.4. Diagnosing the disturbance type on the basis of diagnostic rules

On the basis of the feature separability analysis, Figure 6 demonstrated that the separability of features in deforestation cases is significantly greater than that in fire cases for the selected samples. Therefore, it is necessary to analyse each disturbance type independently to develop effective decision-making rules. Both the DVI and the NDVI exhibit strong and consistent separability across the two cases; however, they fail to distinguish between disturbance types. In fire cases, the CIV of the NBR is lower than that of the PBI and SR, indicating better separability (Figure 6). Conversely, in deforestation cases, the SR has a smaller CIV than the NBR and PBI do, enabling better differentiation of deforestation disturbances. Therefore, in this study, the NBR and SR were selected as the indices for diagnosing forest disturbance types. The change rates (C) of the NBR and SR values for the selected samples were calculated and normalised, and the results

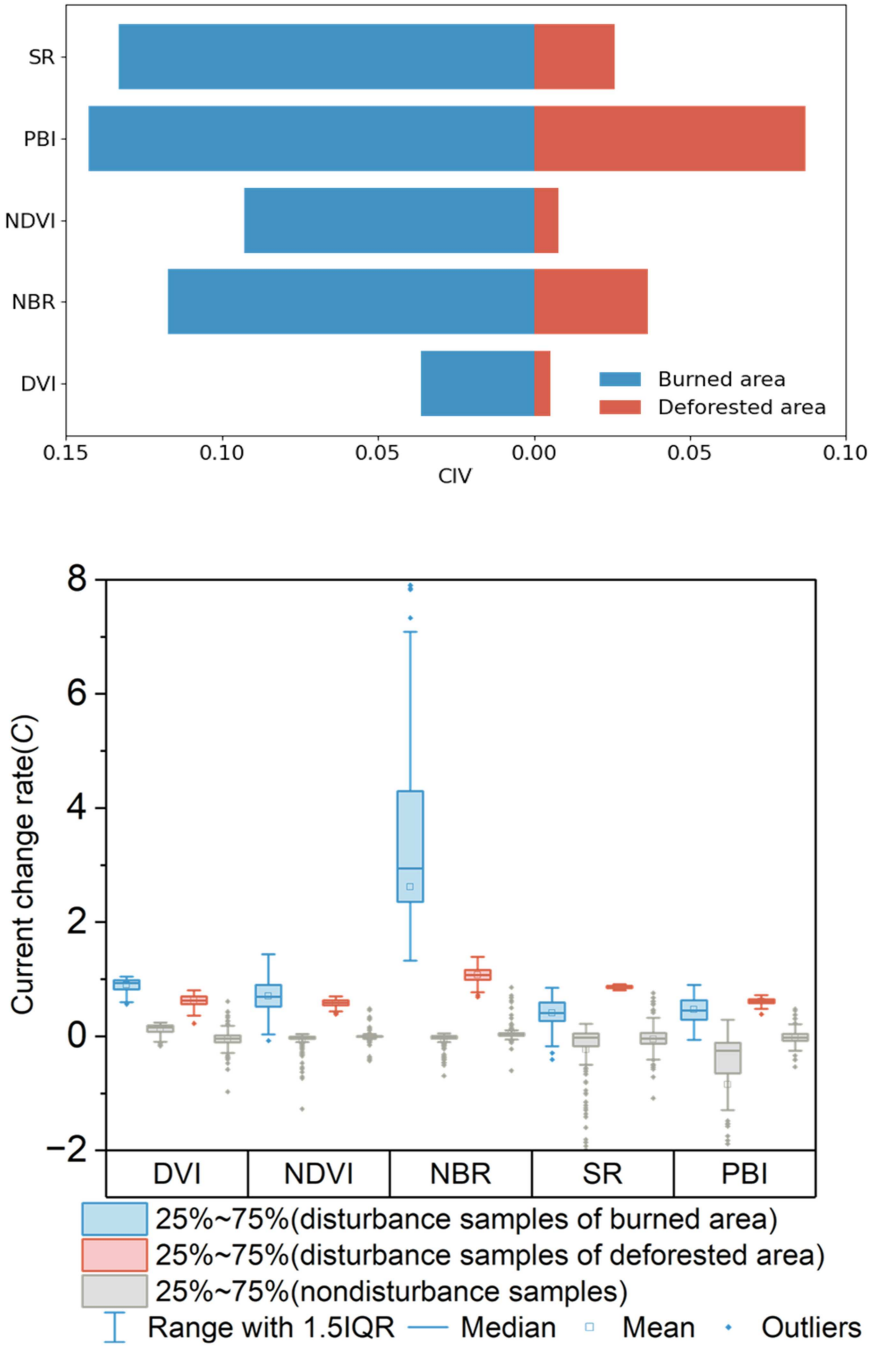


Figure 6. CIV separability of Index features for Italy forest fire and Brazil deforestation.

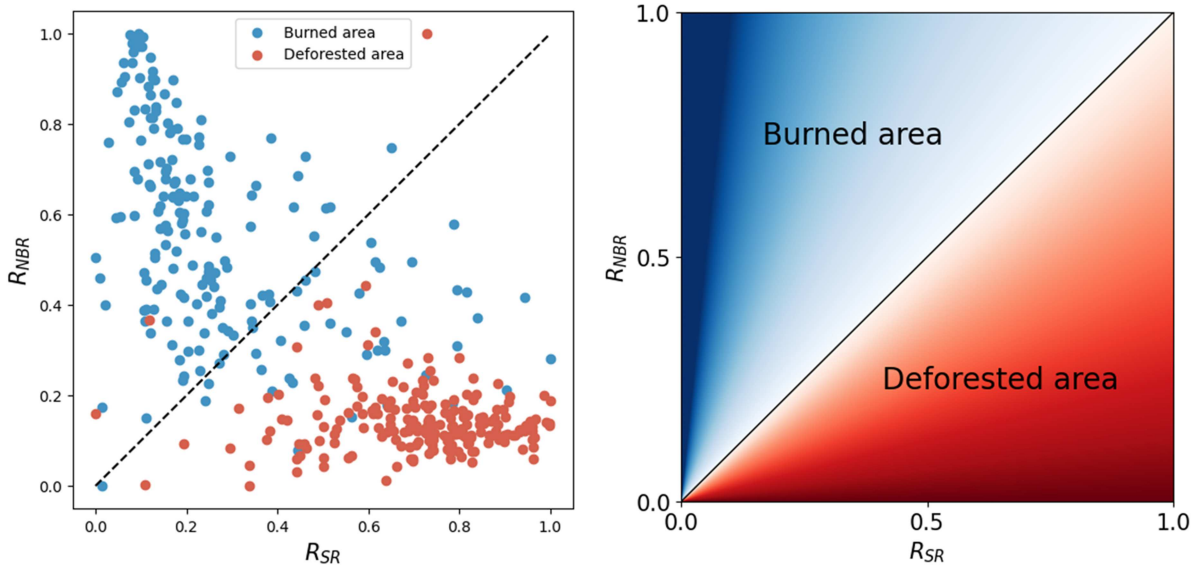


Figure 7. Forest disturbance type diagnostic diagram (blue area represents fire disturbance, red area represents deforestation disturbance, darker colour indicates a higher likelihood of belonging to a certain disturbance type).

are presented in Figure 7. The rate of change in the NBR is typically higher than that in the SR for forest fire events, whereas the SR shows a higher rate of change than the NBR does for deforestation events.

We assume that only one type of disturbance occurs within a given area and diagnose the type of disturbance through the following steps. First, the interquartile range (IQR) method was used to remove outliers in each feature of the disturbance area, with the upper and lower bounds defined as Q3 + 1.5 * IQR and Q1-1.5 * IQR, respectively, to eliminate errors caused by misclassification. Then, a linear normalisation method was applied to scale the valid data of two features to a range between 0 and 1, ensuring that the ratio between the two features was comparable. Finally, the ratio of the NBR to SR ($\Delta R = R_{NBR}/R_{SR}$) was calculated. When ΔR is greater than 1, the disturbance pixel is considered a burned pixel, and the higher the ΔR value is, the greater the likelihood of a fire (represented by the blue area in Figure 7). When ΔR is less than 1, the pixel is considered a deforestation pixel, and the closer ΔR is to 0, the greater the likelihood of deforestation (represented by the red area in Figure 7). The number of pixels with ΔR greater than 1 ($\text{num}_{(\Delta R>1)}$) and the number of pixels with ΔR less than 1 ($\text{num}_{(\Delta R<1)}$) are counted. If $\text{num}_{(\Delta R>1)} > \text{num}_{(\Delta R<1)}$, the area is considered to be affected by fire, and all affected pixels are marked as burned pixels. Otherwise, they are marked as deforestation pixels.

3.3. Result evaluation

3.3.1. Qualitative comparison with public datasets

To qualitatively evaluate the reliability and credibility of the disturbance detection results, a consistency analysis was conducted using publicly available reference datasets. Two widely recognised global disturbance products were selected for cross-validation: the MODIS Burned Area Product (MCD64A1) and the Hansen Global Forest Change (GFC) dataset. These datasets serve as independent references for assessing the spatial and temporal consistency of the detected disturbance areas.

For fire disturbance validation, the MODIS MCD64A1 product was employed. This dataset provides monthly global burned area information at a spatial resolution of 500 m, derived from MODIS satellite imagery. Although its spatial resolution is coarser than that of Sentinel-2 (10 m), it is widely acknowledged for large-scale fire monitoring and serves as a robust benchmark for fire disturbance evaluation. The spatial and temporal correspondence between the detected burned areas and the MODIS product was visually compared. Areas of overlap and temporal coincidence were used as indicators of consistency between the two datasets.

For deforestation validation, the Hansen Global Forest Change (GFC) dataset was used. The GFC dataset provides annual forest loss information derived from Landsat imagery at a 30 m spatial resolution, covering the period from 2000 to 2024. It has been extensively utilised for global forest change assessment. The deforestation regions detected by the proposed algorithm from Sentinel-2 imagery were qualitatively compared with the GFC forest loss regions to assess spatial correspondence and temporal agreement.

Given the differences in spatial resolution, data sources, and temporal coverage between the Sentinel-2–based results and the reference datasets, this comparison focuses on qualitative consistency rather than quantitative accuracy assessment. The visual comparisons aim to confirm whether the detected disturbances correspond to actual fire and deforestation events recorded by independent global datasets.

3.3.2 Quantitative accuracy evaluation based on digital reference datasets

In addition to the qualitative consistency analysis, quantitative accuracy assessment was conducted to evaluate the performance of the automated disturbance detection and diagnostic results. The disturbance reference data were generated through expert visual interpretation of Sentinel-2 imagery following a standardised interpretation protocol. The procedure was independently conducted by two experts to ensure objectivity. The interpretation was based primarily on the current-period image, while the previous-period image and the change rate image were used as auxiliary references to enhance temporal context. Both true-colour and false-colour RGB composites constructed from the selected comprehensive features were used to enhance the visual separability of deforestation and burned areas. Disturbance boundaries were manually delineated along visually evident edges to define the disturbed areas. The two sets of interpretation results were then cross-compared, and any discrepancies were reviewed and resolved by a third expert to reach consensus and ensure the consistency and reliability of the reference data.

Using manual visual interpretation results as a reference, the results of automated forest disturbance detection are evaluated on the basis of overall accuracy (OA), precision (P), recall (R), and the F1 score (F1).

$$OA = \frac{TP + TN}{TP + TN + FP + FN} \tag{8}$$

$$P = \frac{TP}{TP + FP} \tag{9}$$

$$R = \frac{TP}{TP + FN} \tag{10}$$

$$F1 = \frac{2 \cdot Precision \cdot Recall}{Precision + Recall} \tag{11}$$

where TP represents the number of samples correctly classified as positive when the actual class is positive, FN represents the number of samples classified as negative when the actual class is positive, FP represents the number of samples classified as positive when the actual class is negative, and TN represents the number of samples correctly classified as negative when the actual class is negative. Simultaneously, the overall accuracy (OA) is used to evaluate the forest disturbance diagnostic results.

3.3.3 Comparison with recognition results based on other classifiers

To comprehensively evaluate the feasibility of automated disturbance samples extraction and the performance of the OC-SVM algorithm in forest disturbance detection, two widely used supervised classification algorithms—Random Forest (RF) and Support Vector Machine (SVM)—were employed for comparative analysis. Their basic principles and configurations are described as follows:

RF is an ensemble learning method that builds multiple decision trees and combines their predictions through voting (for classification) or averaging (for regression) to improve model accuracy and robustness. The RF algorithm demonstrates strong robustness and generalisation capability, particularly in handling high-dimensional data and datasets with outliers or noise. Additionally, RF offers relatively fast training and prediction speeds, making it suitable for large-scale data processing. However, due to its complex

model structure, it may not perform optimally on imbalanced datasets and often requires considerable

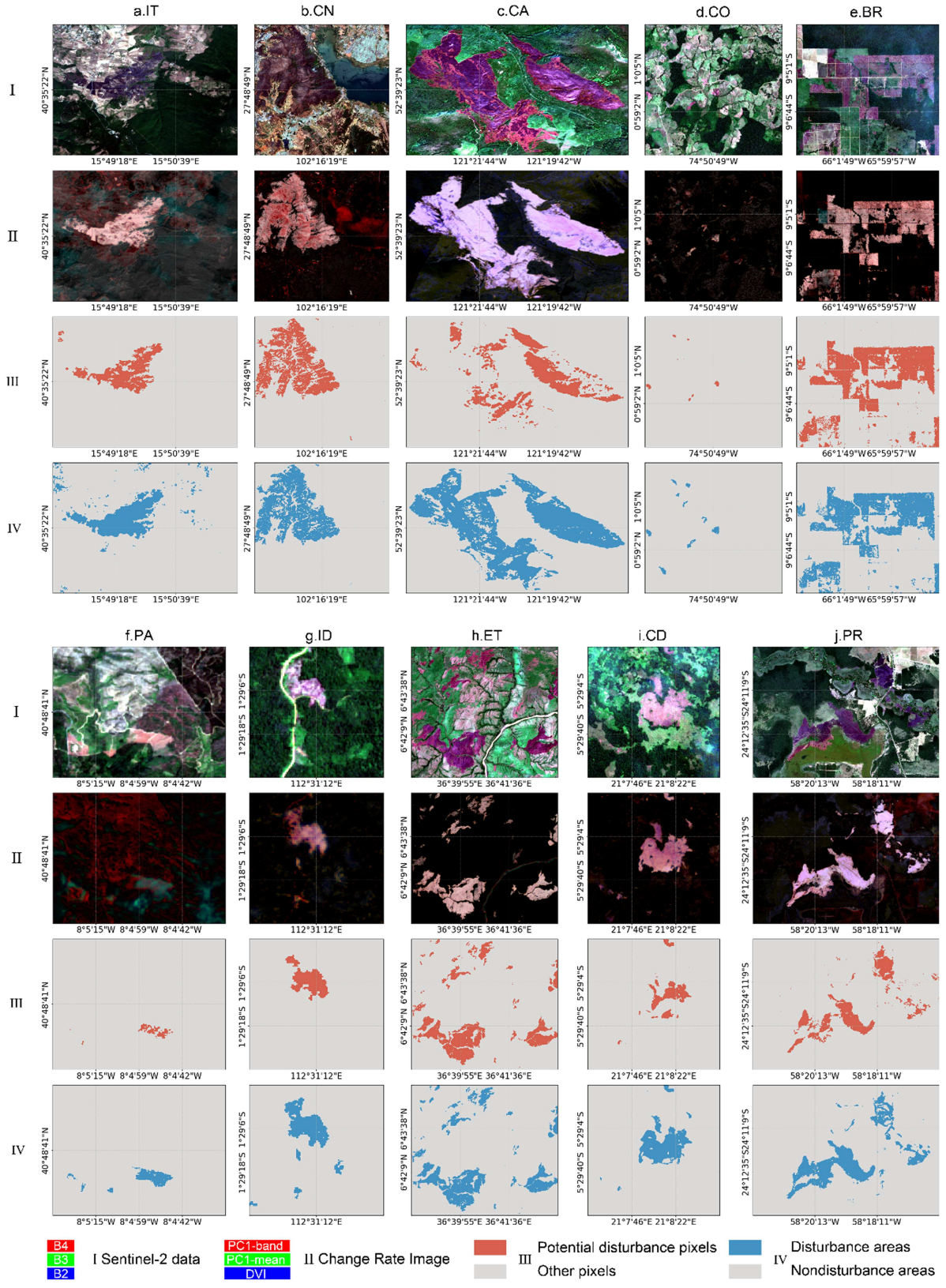


Figure 8. Automated forest disturbance detection results of 10 cases. (I) Post disturbance sentinel-2 images, (II) feature change rate image, (III) potential disturbance pixels, (IV) forest disturbance detection results.

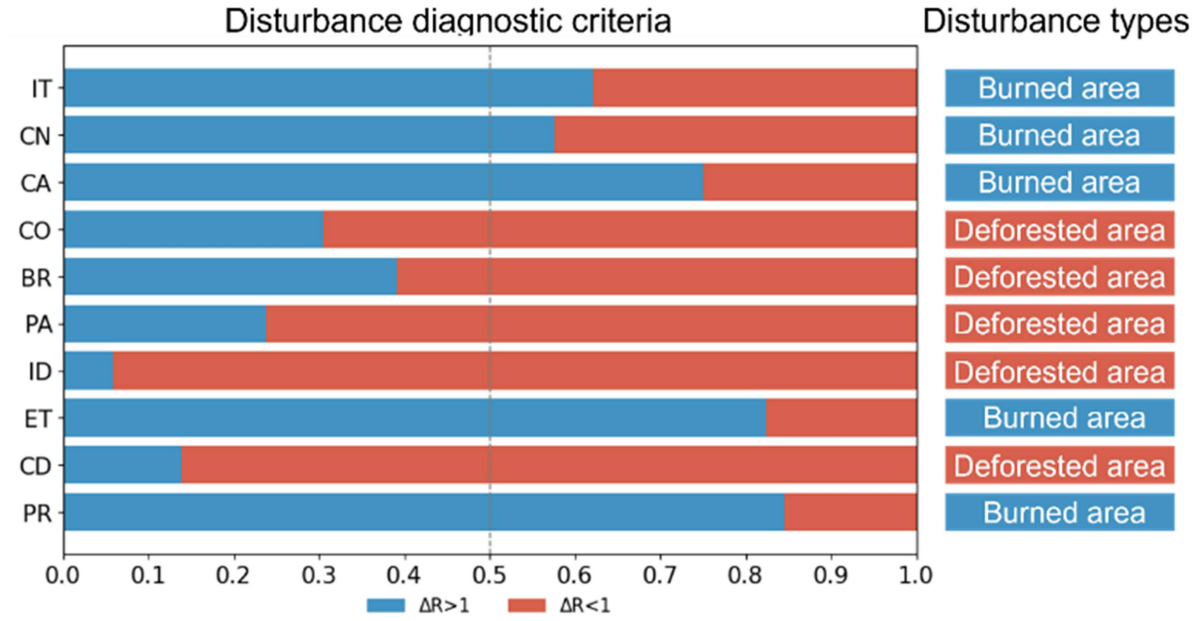


Figure 9. Forest disturbance type diagnosis results.

computational resources to build a large number of trees, which can lead to high computational costs when dealing with very large datasets.

SVM is a supervised machine learning method based on statistical learning theory. The core idea of SVM is to construct an optimal separating hyperplane that maximises the margin between different classes, thereby improving the generalisation ability. The SVM algorithm performs well in scenarios involving small samples and high-dimensional data, effectively reducing the risk of overfitting. However, SVM has relatively high computational complexity. When applied to multi-class classification tasks and large datasets, it often suffers from slow training speed, sensitivity to the choice of kernel functions and hyperparameters, and increased difficulty in parameter tuning.

Both RF and SVM classifiers were trained on the same disturbance input pixels and reference samples as the OC-SVM model, supplemented with representative non-disturbance pixels. Their outputs were subsequently evaluated using the same accuracy metrics, including overall accuracy (OA), precision (*P*), recall (R), and the F1 score (F1).

3.3.4 Comparison with recognition results based on single-phase data

To analyse whether the change rate imagery contributes to improving the accuracy of disturbance extraction, a comparative experiment was conducted using two types of input data: the change rate imagery (*C*) derived from pre- and post-disturbance Sentinel-2 composites, and the single-period imagery (*I*) acquired during the post-disturbance period. Both datasets were constructed using the same set of selected comprehensive features to ensure comparability.

In the change rate imagery (C), each feature represents the pixel-wise difference between the pre- and post-disturbance periods, capturing the relative change in features. In contrast, the single-period imagery (*I*) contains only the feature values of the post-disturbance image. The two datasets were independently input into the OC-SVM model for disturbance detection. All other model parameters and processing steps were kept consistent to ensure that performance differences were solely attributed to the data type.

The detected disturbance maps obtained from both input types were evaluated using the same expert-interpreted reference data. Quantitative accuracy metrics, including overall accuracy (OA), precision (*P*), recall (R), and F1 score (F1), were used to assess and compare the extraction performance.

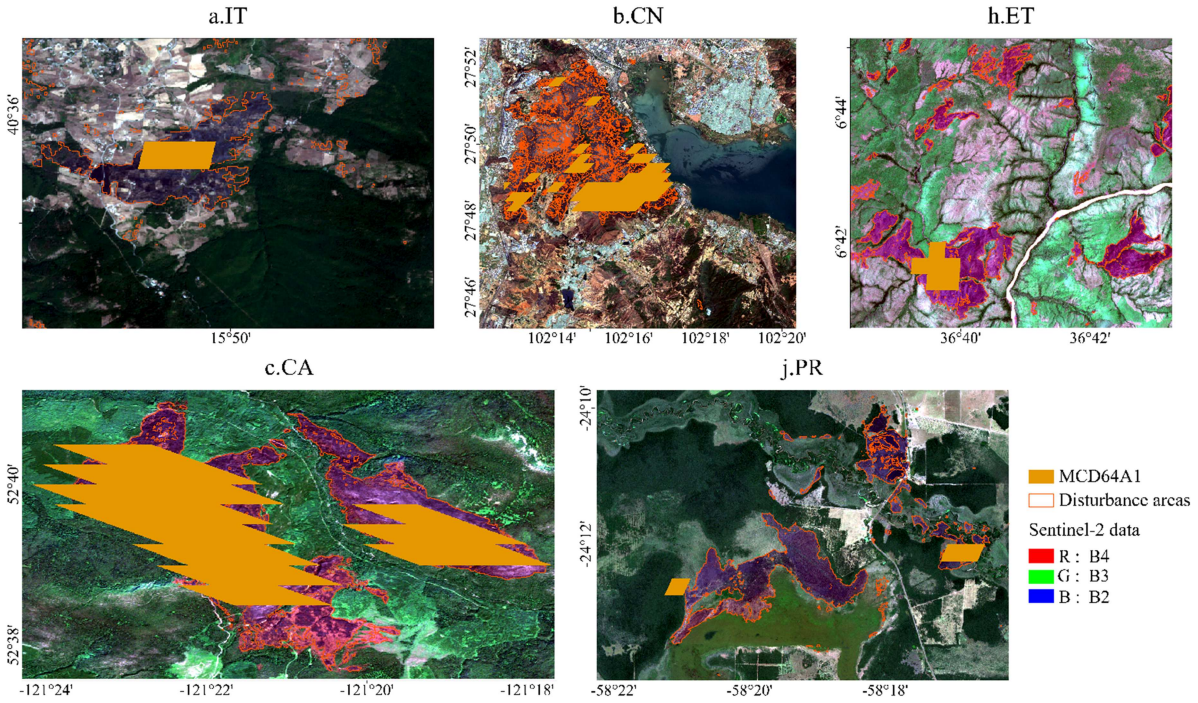


Figure 10. Comparison between the MODIS burned area product (MCD64A1) and the Sentinel-2 based burned area disturbance detection results in the burned study areas.

4. Results

Figure 8 presents the forest disturbance detection results obtained by the proposed automated algorithm across ten representative cases, including both fire and deforestation disturbances distributed in different regions worldwide. From top to bottom, the post-disturbance Sentinel-2 true-colour images (I), the change-rate images derived from the selected comprehensive features (II), the potential disturbance pixel identification results (III), and the final disturbance detection results (IV).

As shown in Figure 8, the change rate imagery effectively enhances the contrast between disturbed and undisturbed areas, allowing the OC-SVM model to detect potential disturbance pixels with high sensitivity. The detected disturbance areas (blue) correspond well with visually interpretable disturbance features, while nondisturbance regions (grey) are effectively excluded. Across all cases, the algorithm successfully delineated the spatial extent of both burned and deforested areas, accurately capturing disturbance boundaries under varying ecological and geographic conditions.

Figure 9 illustrates the diagnostic results of disturbance types for the ten study cases. The diagnostic rule was established based on the change rate (ΔR) characteristics derived from the selected features. Pixels with $\Delta R > 1$ were diagnosed as burned areas, while those with $\Delta R < 1$ were classified as deforested areas. The stacked bars show the proportion of pixels with different change rate values (ΔR). Among the ten cases, IT, CN, CA, ET, and PR were diagnosed as burned areas, while CO, BR, PA, ID, and CD are diagnosed as deforested areas.

Overall, these results demonstrate that the proposed method robustly detects forest disturbance areas and accurately diagnoses their types from Sentinel-2 imagery with high spatial precision. The incorporation of change rate features provides enhanced separability between disturbance and stable forest pixels, supporting accurate disturbance detection and diagnosis. The subsequent sections provide a detailed qualitative and quantitative evaluation of the detection and diagnostic results to further verify their reliability and performance.

4.1. Qualitative validation based on public disturbance datasets

Figure 10 presents a visual comparison between the MODIS burned area and the disturbance detection results. The red outlines represent the detected burned regions from our study, while the yellow areas indicate MODIS burned areas. The comparison shows that the fire disturbance areas detected by our algorithm correspond well with the MODIS burned area data in both spatial distribution and occurrence time. In all fire disturbance cases, the detected burned areas were located within or adjacent to the MODIS burned regions during the corresponding period, confirming the temporal and spatial consistency between the two datasets. This agreement demonstrates that the detected fire disturbances are consistent with actual fire events captured by the MODIS product.

Figure 11 presents a visual comparison between the deforestation detection results and the GFC forest loss dataset. The red outlines indicate the deforestation areas detected by our algorithm, while the yellow regions represent forest loss areas from the GFC product. In the BR, ID, and PA cases, the detected disturbance patches fall within or overlap substantially with the GFC-identified loss areas, indicating strong spatial consistency between the two datasets. In the CO and CD cases, the GFC dataset failed to capture certain deforestation regions that were clearly identified by our algorithm. Based on expert visual interpretation of Sentinel-2 imagery, these areas indeed correspond to actual deforestation events that were not recorded in the GFC product, demonstrating the superior sensitivity of the proposed method in detecting small-scale or short-term deforestation disturbances.

These qualitative comparisons confirm that the disturbance areas detected by the proposed algorithm are highly consistent with widely recognised public disturbance datasets in both spatial distribution and temporal occurrence. The results validate the robustness and reliability of the method in accurately detecting and distinguishing forest disturbance events across different disturbance types and geographic regions.

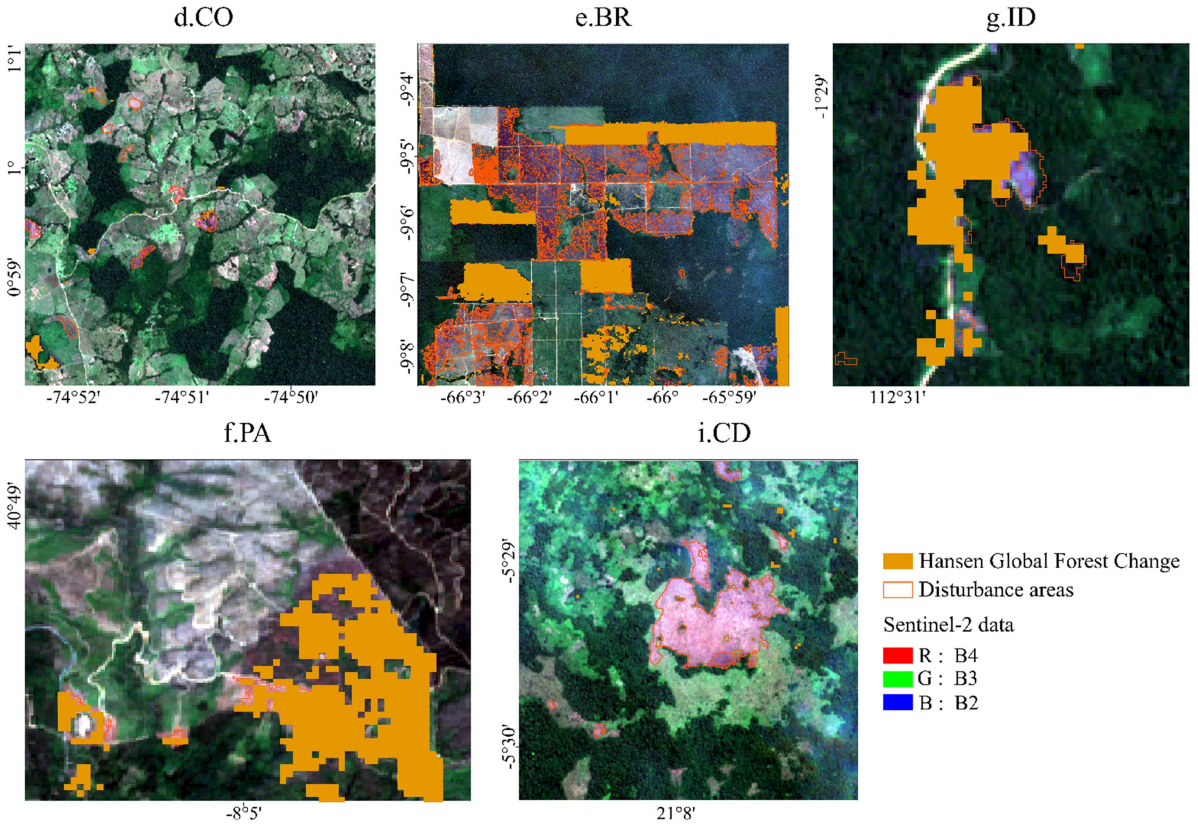


Figure 11. Comparison between the GFC and the Sentinel-2 based deforestation detection results in the deforested study areas.

4.2. Quantitative validation based on reference data

The accuracy evaluation results of the forest disturbance detection results are presented in Figure 12. The average overall accuracy (OA), precision (*P*), recall (R), and F1 score (F1) across the ten cases were 98.46%, 95.78%, 89.05%, and 0.923, respectively. Among all cases, the highest overall accuracy (99.80%) was achieved for CO, while the highest F1 score of 0.979 was achieved for CD. The results indicate that the proposed algorithm achieved consistently high detection performance across diverse ecological and geographic conditions.

For disturbance type diagnosis, the results were fully consistent with the verified event records, achieving a diagnostic accuracy of 100%. Specifically, in deforestation cases, the average OA and F1 score respectively reached 98.40% and 0.917, while in fire cases, the corresponding values were 98.51% and 0.928.

The automated disturbance detection algorithm demonstrated excellent quantitative performance in both detecting and diagnosing forest disturbances. The selected comprehensive disturbance features effectively captured forest disturbance information and showed good generalisability within the tested

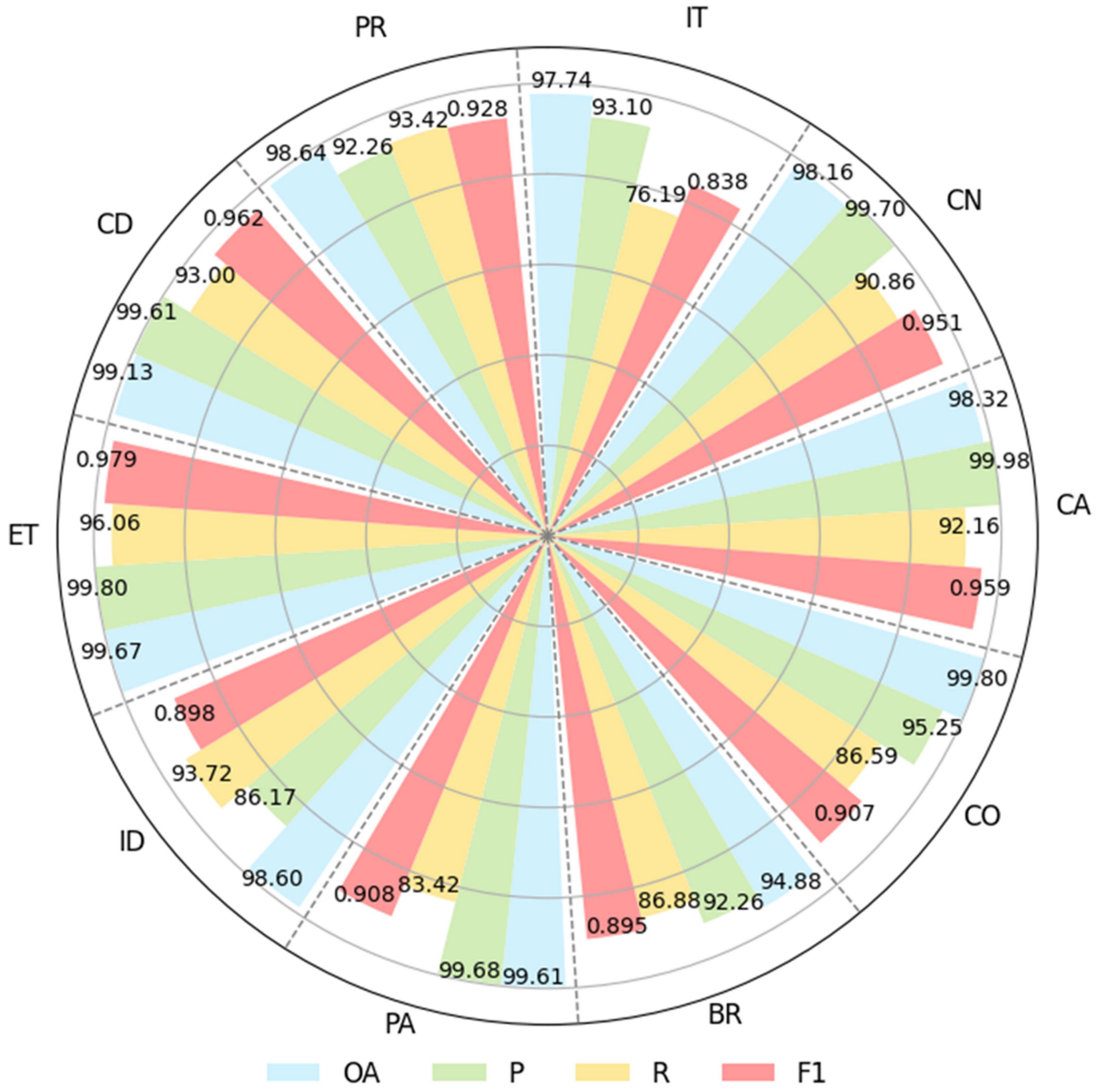


Figure 12. Accuracy evaluation for 10 forest disturbance cases.

cases. With the integration of change rate features and OC-SVM classification, the algorithm achieved precise delineation of disturbance areas and reliable identification of disturbance types, confirming its robustness and reliability for forest disturbance monitoring.

4.3. Comparative validation with other classifiers (RF and SVM)

Figure 13 presents the comparative validation results of disturbance detection using the proposed OC-SVM, RF, and SVM classifiers across ten representative cases. The results show that the OC-SVM algorithm achieved the most accurate and stable disturbance delineation among the three methods. The disturbance patches detected by OC-SVM (I) closely match the visually interpretable disturbance areas, while those produced by RF and SVM (II and III) show overestimation or omission in some regions.

Figure 14 presents the quantitative detection accuracy of the OC-SVM, RF, and SVM algorithms. The OC-SVM algorithm exhibits the best overall detection performance. Specifically, the average OA of OC-

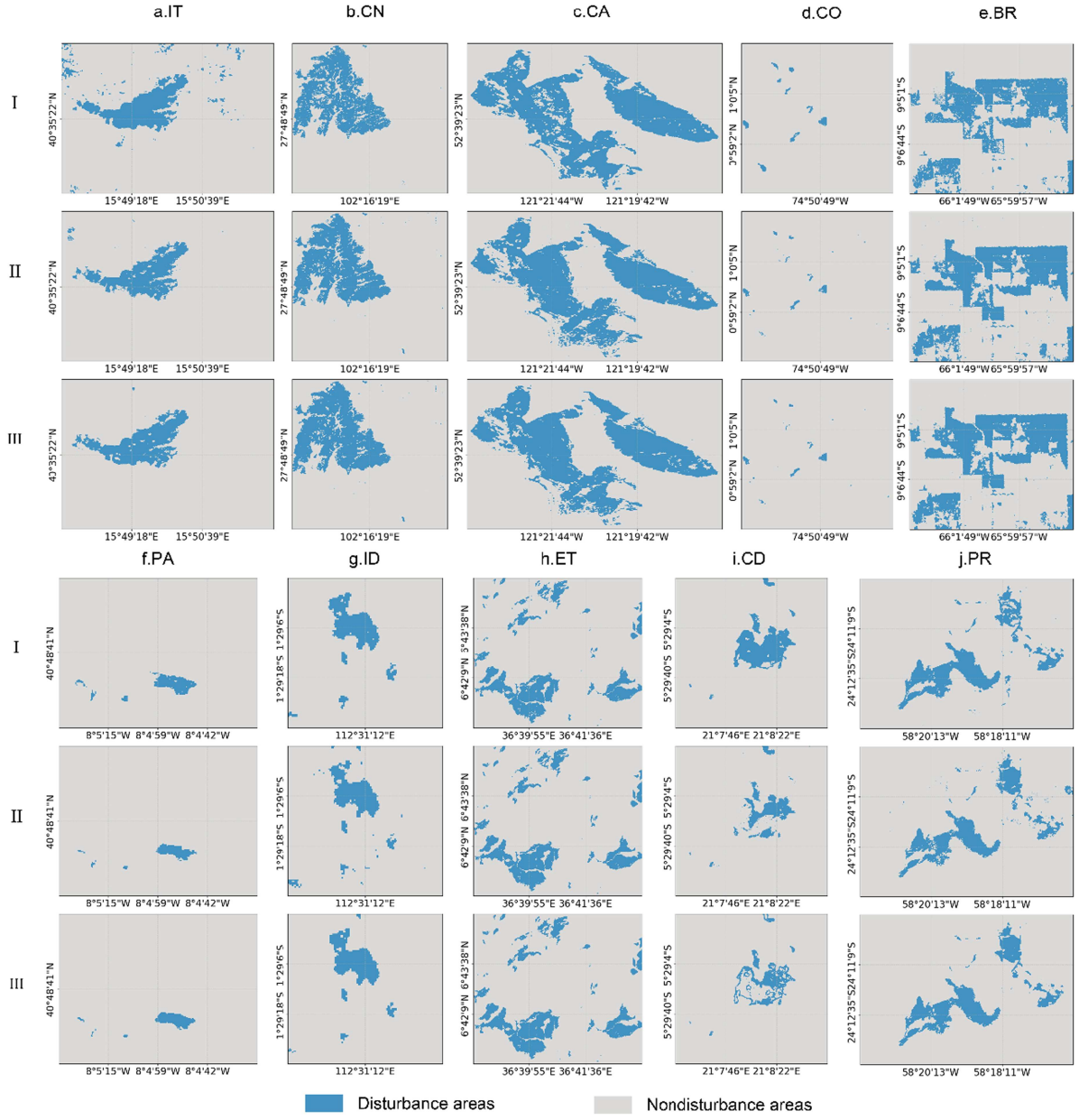


Figure 13. Detection results of forest disturbance of 3 methods on 10 cases. (I) OC-SVM, (II) RF, (III) SVM.

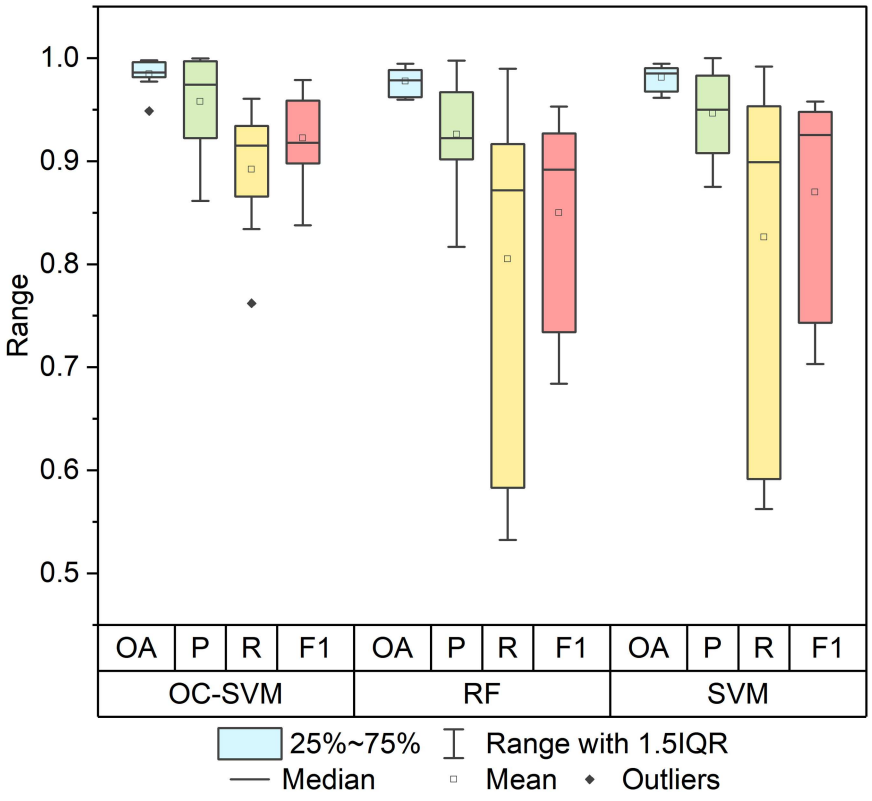


Figure 14. Comparison of forest disturbance detection accuracy among three methods.

SVM is 98.46%, with an average *P* of 95.78%, an average R of 89.23%, and an average F1 of 0.923. The RF algorithm achieves an average OA of 97.77%, an average *P* of 92.63%, an average R of 80.51%, and an average F1 of 0.850, whereas the corresponding values of the SVM algorithm are 98.14%, 94.66%, 82.66%, and 0.870, respectively.

To further evaluate the performance of the three algorithms, the means, medians, and interquartile ranges of each evaluation metric were computed across all cases (Figure 15). The OC-SVM shows higher

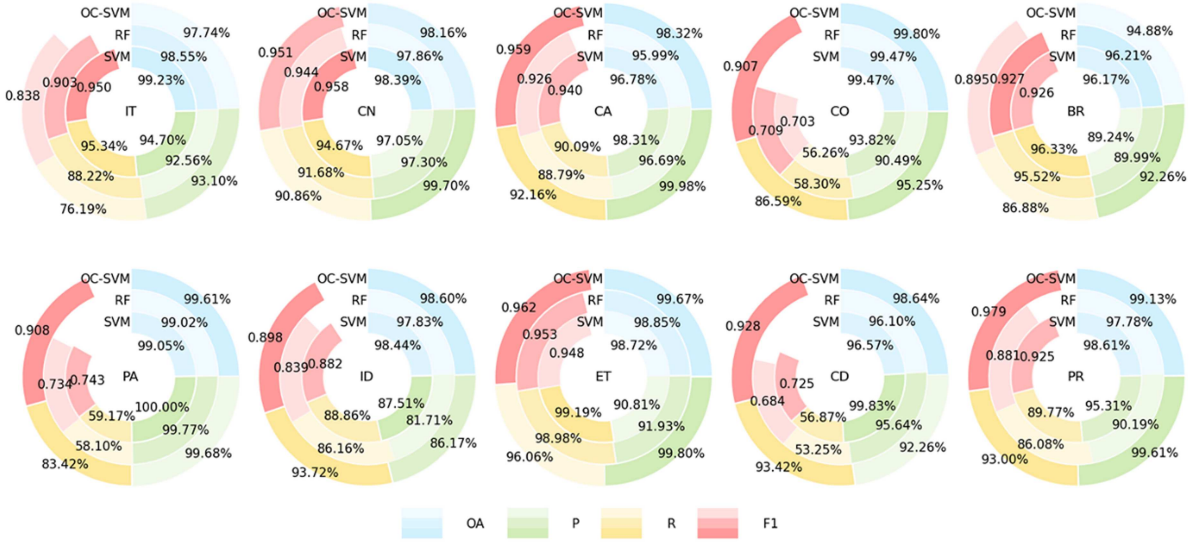


Figure 15. Accuracy evaluation of three classifiers on 10 forest disturbance cases.

median values for OA, *P*, R, and F1, as well as narrower interquartile ranges, reflecting stronger stability and reliability. This finding suggests that the OC-SVM algorithm can provide consistent and excellent results for detection tasks. In contrast, the RF and SVM algorithms result in wider distributions of R and F1, indicating significant fluctuations in detection performance across different cases.

An analysis of accuracy differences in specific cases reveals that for the IT, CN, CA, BR, ID, ET, and PR cases, there is no significant difference in detection accuracy among the three algorithms, indicating high robustness in these scenarios. However, in the CO, PA, and CD cases, the OC-SVM detection results are significantly better than those of the RF and SVM methods. The comparative results confirm that the proposed OC-SVM classifier outperforms RF and SVM in both accuracy and stability. It provides more precise and consistent disturbance delineation, demonstrating superior adaptability to diverse disturbance conditions. Further examination of the historical images and threshold maps for these cases indicated that some of the preprocessed historical images were still affected by residual clouds. Although cloud masking was applied, a small number of cloud-contaminated pixels remained, resulting in abnormally high threshold values in the historical change rate images. Consequently, even true disturbance pixels might have been mistakenly labelled as non-disturbance samples due to their elevated thresholds, and were then used as training samples for the RF and SVM classifiers. These mislabeled samples degraded the classification accuracy of RF and SVM.

In contrast, the OC-SVM model requires only disturbance samples for training, thereby reducing the risk of accuracy degradation caused by mislabeled non-disturbance samples. Although the residual clouds also reduced the number of input disturbance samples for the OC-SVM, the model still successfully identified more disturbance pixels during classification. This result demonstrates that the OC-SVM effectively captured the intrinsic characteristics of disturbance samples and exhibited stronger generalisation capability in cloud-affected scenarios, leading to more accurate and stable disturbance detection results.

4.4. Comparative validation between change rate and single-period imagery

As shown in Figure 16, the disturbances extracted from change rate imagery are markedly more accurate than those derived from single-period imagery. The average overall accuracy (OA) of detection increased from 87.75% with single-period imagery to 98.46% with change rate imagery, while the F1 score improved by 0.214. The use of change rate imagery substantially enhances the comparability between images by emphasising temporal variations, thereby facilitating the identification of anomalous changes. This improvement strengthens the foundation for disturbance pixel extraction, making disturbance regions

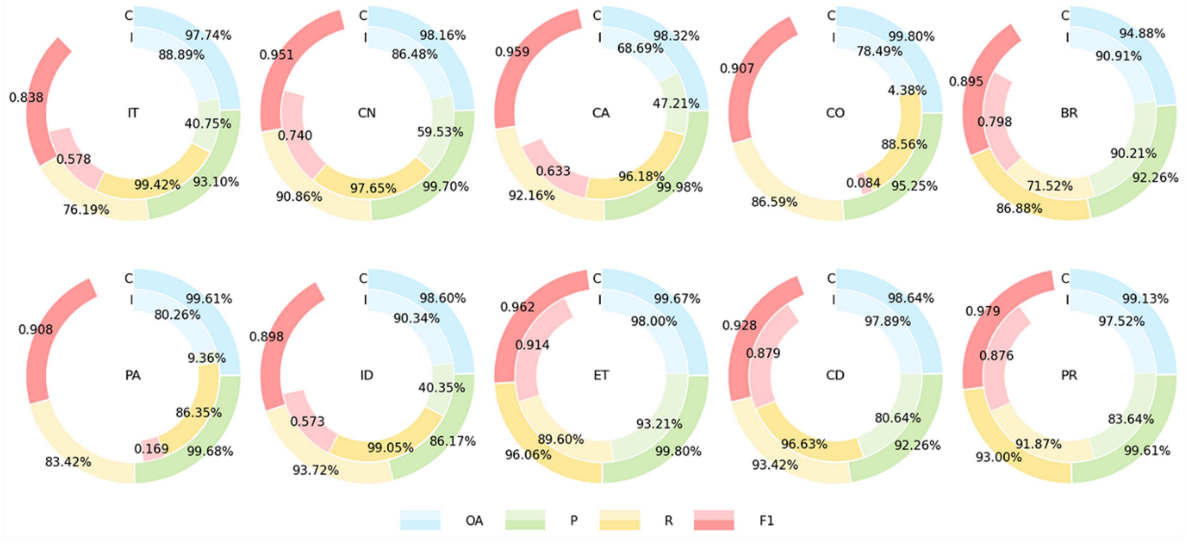


Figure 16. Accuracy evaluation of change rate image (*C*) and single-period image (*I*) on 10 forest disturbance cases.

more prominent and easily distinguishable from stable forest areas. In forest disturbance monitoring, the change rate imagery effectively highlights significant alterations induced by natural disturbances or anthropogenic activities, providing clearer visual cues and yielding superior detection performance. Overall, these findings demonstrate that incorporating change rate features significantly improves the precision and reliability of automated forest disturbance detection.

5. Discussion

Forest disturbances such as fire and deforestation lead to significant changes in canopy structure, chlorophyll content, and moisture conditions, which directly affect spectral reflectance, particularly in the red, near-infra-red (NIR), and red-edge regions. Sentinel-2 B6 (red-edge 1), B7 (red-edge 2), B8 (NIR), and B8A (red-edge 3) are highly sensitive to these biophysical changes, making them effective indicators for detecting disturbance-induced spectral anomalies. The mean texture features derived from these bands capture spatial heterogeneity and canopy fragmentation caused by disturbances, providing complementary structural information beyond spectral intensity. The Difference Vegetation Index (DVI), which emphasises variations between NIR and red reflectance, enhances the contrast between healthy and disturbed vegetation, allowing for robust detection of vegetation loss or degradation.

Methodologically, the proposed approach integrates physical thresholding and machine learning in a complementary manner, enhancing both detection reliability and generalisation. The initial threshold-based step effectively identifies potential disturbance regions by leveraging the physical response of vegetation change in the spectral domain, while the subsequent OC-SVM refinement compensates for the limitations of thresholding when residual noise or cloud contamination is present in historical reference imagery. Residual clouds, even after masking, can elevate threshold values in historical change rate images, resulting in missed detections of true disturbances. By learning from disturbance samples with clear spectral and structural characteristics, the OC-SVM effectively redefines the decision boundary in feature space, recovering genuine disturbance pixels that might have been excluded by overly conservative thresholds.

This combined strategy reduces both omission errors due to incomplete threshold extraction and commission errors caused by classifier overfitting. Unlike traditional supervised methods such as RF and SVM, which require accurately labelled disturbance and nondisturbance samples, the OC-SVM only depends on disturbance samples and is therefore less affected by mislabeled data or residual background noise. This makes the method particularly robust under heterogeneous and cloud-prone environmental conditions, where reliable nondisturbance samples are difficult to obtain. The results confirm that the OC-SVM captures the intrinsic feature distribution of disturbance events and generalises well across diverse geographic regions, leading to more accurate and stable detection outcomes. Overall, the integration of physically interpretable thresholds with a one-class learning framework ensures both methodological soundness and operational adaptability, particularly for automated forest disturbance monitoring.

The observed consistencies and discrepancies between the proposed disturbance detection results and the public disturbance datasets (MCD64A1 and GFC) can be explained by the differences in data sources, temporal resolution, and detection strategies. The higher spatial resolution of Sentinel-2 imagery and the use of short-term change rate images enable the proposed algorithm to capture fine-scale disturbance patterns that emerge immediately after disturbance events. This approach highlights localised and abrupt spectral transitions that are often blurred or missed in coarser datasets. In contrast, the coarse spatial resolution of MODIS imagery (500 m) tends to obscure small burned areas and introduce mixed-pixel effects, particularly in heterogeneous forest landscapes. Similarly, the annual temporal scale of the GFC dataset, which prioritises global coverage and cloud-free composites, inevitably suppresses short-term or within-year forest loss signals. Consequently, deforestation events that occur over a short duration or are quickly followed by vegetation regrowth may be omitted from the annual composite. Overall, these differences suggest that the proposed method, benefiting from high-resolution Sentinel-2 data and change rate imagery, is inherently more sensitive to subtle or short-term forest disturbances than global products designed for long-term and large-scale monitoring.

While the use of change rate imagery substantially improves disturbance detection accuracy, several limitations should be acknowledged. Change rate image inherently captures only the variations that

occurred between two specific image dates, thus reflecting disturbances that took place within that interval rather than representing the absolute condition of any single period. To ensure reliable and stable detection, this study used monthly median composites rather than higher-frequency images. Although Sentinel-2 provides a 5-day revisit capability, monthly compositing effectively reduces the influence of cloud contamination, atmospheric noise, and data gaps, particularly in tropical regions. At the same time, using a longer temporal interval between image pairs makes seasonal vegetation changes more pronounced, broadening the threshold range and helping to distinguish true disturbances from phenological fluctuations. The inclusion of one year of historical data further aligns the dynamic thresholds with recent vegetation conditions and prevents overfitting to outdated baselines.

Despite the promising results, some challenges remain. The current method focuses primarily on short-term disturbances and has not yet been fully validated for long-term monitoring, where vegetation regrowth or post-disturbance recovery may lead to classification uncertainty. Moreover, the assumption that each disturbed area corresponds to a single disturbance type may not hold in complex ecosystems where multiple disturbance processes—such as fire followed by logging—occur sequentially or concurrently. In addition, the experimental validation in this study was conducted using a limited number of regional-scale cases rather than large-scale or global assessments. As a result, the generalisation of the algorithm to diverse forest types, climatic zones, and biogeographic contexts remains to be further verified. Future research should therefore expand the spatial scope of testing and integrate multi-temporal sequence analysis, probabilistic modelling, and multi-label classification frameworks to better capture overlapping disturbance dynamics and improve the interpretability of automated disturbance detection.

6. Conclusions

In this study, we developed and validated an automated forest disturbance monitoring algorithm based on Sentinel-2 data, aimed at achieving unsupervised, highly adaptive, and accurate disturbance range detection and type diagnosis for forest fires and deforestation. The results show that 1) the comprehensive features selected through the CIV and JM distances are effective for detecting fire burned areas and deforestation disturbances. The unique red-edge bands, near-infra-red bands, corresponding mean texture features, and DVI of Sentinel-2 make significant contributions to the detection of both forest fires and deforestation. 2) The dynamic threshold method, which is based on historical change rate images, can detect disturbance samples precisely and automatically. 3) The use of the OC-SVM algorithm to detect disturbance areas in ten cases yielded an average OA of 98.46% and an average F1 of 0.923. 4) The proposed disturbance type diagnosis method achieved 100% diagnostic accuracy in all ten cases. In conclusion, our method has high practical value for complex, diverse forest disturbance monitoring tasks and has the potential to be expanded to other disturbance type monitoring tasks.

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Author contributions

CRediT: **Xiufang Zhu:** Methodology, Conceptualisation; **Junying Song:** Data curation, Writing – review & editing, Writing – original draft; **Mingxiu Tang:** Writing – review & editing, Resources; **Rui Guo:** Writing – review & editing.

Disclosure statement

The authors report there are no competing interests to declare.

Data availability statement

The data that support the findings of this study are available from the corresponding author, upon reasonable request.

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